The AI Revolution in Biology: Decoding Life with Algorithms

From protein synthesis in high school Biology classes to AI-powered drug discovery and AlphaFold, this project explores how Artificial Intelligence is changing the way we understand life.

Prepared by

Sarp AKAR

Prospective Aerospace Engineering Student

Interested in Science, Technology, Engineering, and Research

Scientific Mentor / Supervisor

Okan Kaan Binzat

Biology Teacher

ODTÜ Development Foundation Schools

Research Question

How are Artificial Intelligence and Machine Learning transforming modern Biology, and how are these technologies accelerating scientific discovery and personalized healthcare?

Focus Areas

Artificial Intelligence

Molecular Biology

Bioinformatics

Computational Biology

Personalized Medicine

Date

Sept 2026

Ankara, Türkiye

1. THE PROTEIN FOLDING PUZZLE AND THE ALPHAFOLD REVOLUTION

1.1. Biology's Greatest Mystery: 3D Protein Folding

One of the first things we learn in Biology is that DNA is transcribed into RNA, and RNA is then translated into proteins. Simple enough, right?

What fascinated me was what happens next.

Once a protein is produced, it doesn't stay as a long chain of amino acids. Within seconds, it twists, bends, and folds into a complex three-dimensional structure. That structure is what determines how the protein actually works inside a living organism.

For decades, scientists tried to predict how a protein would fold just by looking at its amino acid sequence. This challenge became known as the Protein Folding Problem.

The problem was incredibly difficult. Traditional methods such as X-ray crystallography and cryo-electron microscopy could take months or even years to determine the structure of a single protein, often at very high cost. For years, this remained one of Biology's biggest unsolved puzzles.

1.2. Enter Google DeepMind and AlphaFold

Everything changed when Google DeepMind introduced AlphaFold.

Using a Deep Learning model trained on known protein structures, AlphaFold became capable of predicting the three-dimensional shape of proteins with remarkable accuracy and in a fraction of the time required by traditional methods.

What impressed me most was the scale of the achievement.

Instead of solving one protein structure at a time, AlphaFold helped generate structural predictions for nearly all known proteins, creating a database containing information on approximately 200 million proteins and making it freely available to researchers around the world.

As a student interested in science and technology, exploring this project showed me how quickly modern research is changing. Problems that once required years of laboratory work can now be approached using powerful computational tools and Artificial Intelligence.

For me, AlphaFold was one of the clearest examples of how data, algorithms, and scientific research.

2. GENERATIVE AI AND DE NOVO PROTEIN DESIGN

2.1. Hacking Nature: Designing Proteins from Scratch

At this point, I started asking myself another question:

If AI can predict protein structures, can it also create completely new proteins?

The answer surprised me.

Just like ChatGPT can generate sentences that have never existed before, Generative AI models can now design proteins that have never existed in nature. Scientists call this process De Novo Protein Design.

One of the most interesting examples I came across was RFdiffusion, a model developed by David Baker and his team. What amazed me was the idea that a computer could design an entirely new protein for a specific task.

Researchers are already exploring ways to create proteins that can break down plastic pollution, target cancer cells, or perform functions that natural proteins cannot easily achieve.

For me, this was one of those moments where science started to feel a little like science fiction.

2.2. The Rise of the Digital Laboratory (In Silico Biology)

Another thing that caught my attention was how much biology is changing.

For a long time, researchers mainly worked in laboratories using experiments and observations. Today, a growing amount of biology is also happening inside computers.

Scientists call this "In Silico Biology."

Instead of testing every possibility in a laboratory, AI models can simulate millions of molecular combinations in seconds and help researchers focus on the most promising options.

What interested me most was seeing how coding, data, and scientific problem-solving can come together to help researchers understand and model complex systems.

The more I learned about these technologies, the more I realized that modern science is no longer limited to laboratories. Biology, computer science, data analysis, and artificial intelligence are becoming increasingly connected.

Seeing these different fields work together was one of the most exciting parts of this research project.

3. AI-DRIVEN DRUG DISCOVERY AND ECOTOXICOLOGY

3.1. The Biggest Challenge in Drug Development: Time and Cost

Have you ever wondered how long it takes for a new medicine to go from an idea in a laboratory to a product available in a pharmacy?

The answer surprised me.

On average, developing a new drug can take **10 to 12 years** and cost billions of dollars. Researchers must test thousands of different chemical compounds, and most of them fail at some stage of development. Years of work can be lost before scientists finally discover a promising candidate.

This is where Artificial Intelligence is changing the game.

Instead of testing compounds one by one, AI models can analyze the structure of a target protein found in a virus, bacterium, or disease pathway and rapidly search through billions of potential molecules. Through a process known as **Virtual Screening**, researchers can identify the most promising candidates in a fraction of the time required by traditional methods.

What I found most interesting was seeing how computers can now help scientists make decisions that once required years of laboratory testing. Rather than replacing researchers, AI is becoming a powerful tool that helps them work faster, more efficiently, and more accurately.

3.2. From Years to Months: Real-World AI Success Stories

One of the most impressive examples of AI's impact happened in recent years when a company called Insilico Medicine used Artificial Intelligence to identify, design, and advance a potential drug candidate for a completely new disease in only 18 months.

What makes this achievement remarkable is that the early stages of drug discovery traditionally take 5 to 6 years. With AI-assisted tools, researchers were able to reduce that process from years to months.

Another area where AI is making a difference is ecotoxicology. Instead of relying only on animal testing to understand whether a chemical might be harmful to marine life or the environment, AI-powered toxicity models can predict potential effects at the cellular level using computational analysis.

I found this particularly interesting because it shows how technology can help researchers make faster and more informed decisions while also reducing the need for animal testing.

At the same time, these tools may help scientists develop safer and more environmentally friendly solutions by identifying potential risks before substances are released into the environment.

For me, this was another example of how Artificial Intelligence is not only making science faster, but also helping researchers work more efficiently and responsibly.

4. PERSONALIZED HEALTHCARE AND DIGITAL TWINS

4.1. The End of the “One-Size-Fits-All” Medicine Era

In one of my previous projects, I explored pharmacogenomics and learned how genetic differences can influence the way people respond to medications. One of the best examples is the CYP450 family of enzymes in the liver, which helps break down many of the drugs we use every day.

This is where Artificial Intelligence becomes incredibly powerful.

AI systems can combine information from a person's genome, medical records, blood test results, microbiome data, and even data collected from wearable devices. Instead of looking at these pieces separately, AI can analyze them together and identify patterns that would be difficult for humans to detect on their own.

What I found most interesting is the idea that future medicine may become far more personalized than it is today.

Rather than giving the same treatment to everyone, doctors may be able to use AI to develop treatment plans that are tailored to each person's unique biology and medical history.

For me, this was another example of how data, technology, and science are becoming increasingly connected. It also showed me how Artificial Intelligence has the potential to improve both the effectiveness of treatments and the overall patient experience.

4.2. Digital Twins: The Hospitals of the Future

The most futuristic concept I came across during this project was definitely Digital Twin technology.

The basic idea is surprisingly simple: using Artificial Intelligence and large amounts of biological, physiological, and genetic data, researchers can create a virtual version of a patient inside a computer.

What makes this so interesting is what happens next.

Instead of immediately testing a treatment on a real patient, doctors may first be able to test it on that patient's digital twin. AI models can simulate how the body might respond to a medication, treatment, or medical procedure before it is actually used in real life.

The first time I learned about this concept, it sounded like something from a science fiction movie. However, the more I read about it, the more I realized that researchers are already working toward making this idea a reality.

What fascinated me most was seeing how biology, medicine, data science, and Artificial Intelligence can all work together to create something completely new.

For me, Digital Twins represent one of the clearest examples of how technology is changing healthcare. They show how future medicine may become more predictive, personalized, and data-driven than ever before.

5. COMPUTATIONAL THINKING AND THE FUTURE OF SCIENCE

5.1. The End of the "Coding-Free Scientist" Era

One of the biggest things I realized while working on this project is that modern science is changing rapidly.

For a long time, people often imagined biologists spending most of their time in laboratories, using microscopes and conducting experiments. While those skills are still important, today's scientists also need to work with large amounts of data.

A single human genome can generate enormous amounts of information, and researchers increasingly rely on computational tools to analyze, organize, and understand these datasets. Programming languages such as Python and advanced machine learning algorithms are becoming essential tools in many scientific fields.

What I found most interesting is that the boundaries between disciplines are becoming less clear. Biology, computer science, data analysis, and Artificial Intelligence are no longer separate worlds. They are increasingly working together to solve complex problems.

This project helped me better understand how data, technology, and scientific thinking can be combined to accelerate discovery and innovation.

As I learned more about Artificial Intelligence, computational models, and scientific data analysis, I became increasingly interested in how similar approaches are also used in engineering and technology.

For me, one of the most valuable lessons from this project was realizing that understanding data and learning how to work with technology may become just as important as understanding the science itself.

5.2. Career Planning

One of the most valuable things I learned from this project was how closely science, technology, data, and artificial intelligence are connected.

Although this project focused on Biology, many of the concepts I explored, including data analysis, modeling, computational thinking, and problem-solving, are also used in engineering and technology.

Learning about these topics played an important role in helping me discover my growing interest in Aerospace Engineering, research, and technology. It also showed me how different scientific fields can work together to solve complex challenges.

6. Conclusion Report

One of the biggest things I learned from this project is how quickly Biology is becoming a data-driven science.

Before working on this project, I mostly thought of AI as something connected to chatbots, image generation, and everyday technology. However, I discovered that AI is also helping researchers predict protein structures, design new proteins, accelerate drug discovery, and develop more personalized healthcare approaches.

What interested me most was seeing how biology, data, technology, and artificial intelligence are becoming increasingly connected. This project showed me that many scientific challenges can now be explored using both laboratory research and computational tools.

Ultimately, this project helped me better understand how scientific discovery is changing and how different fields can work together to solve complex real-world problems.

7.Personal Reflection

Before starting this project, I mostly associated Artificial Intelligence with chatbots, image generation, and everyday technology.

What surprised me was discovering how important AI is becoming in modern science. Learning about AlphaFold, protein design, drug discovery, and digital twins showed me that AI is helping researchers solve some of the most challenging scientific problems in ways that were almost impossible only a few years ago.

One of the things I enjoyed most about this project was seeing how data, algorithms, and scientific research can work together. It made me realize that many scientific breakthroughs today happen where different disciplines meet.

This project strengthened my interest in Artificial Intelligence, data analysis, and computational thinking. It also helped me better understand how technology can be used to explore, model, and solve complex real-world problems.

Most importantly, this project reminded me that many of today's biggest discoveries start with curiosity, questions, and a willingness to explore new ideas.

8. Skills Developed

Data Analysis

Scientific Research

Problem Solving

Scientific Communication

Technology Literacy

9.References (APA 7)

1. Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., … & Hassabis, D. (2021).
Highly accurate protein structure prediction with AlphaFold.
Nature, 596(7873), 583-589.
https://doi.org/10.1038/s41586-021-03819-2

2. Watson, J. L., Juergens, D., Bennett, N. R., Trippe, B. L., Yim, J., Eisenach, H. E., … & Baker, D. (2023).
De novo design of protein structure and function with RFdiffusion.
Nature, 620(7976), 1089-1100.

3. Stokes, J. M., Yang, K., Swanson, K., Jin, W., Cubillos-Ruiz, A., Donghia, N. M., … & Collins, J. J. (2020).
A deep learning approach to antibiotic discovery.
Cell, 180(4), 688-702.

4. Topol, E. J. (2019).
High-performance medicine: the convergence of human and artificial intelligence.
Nature Medicine, 25(1), 44-56.

5. Rajpurkar, P., Chen, E., Banerjee, O., & Topol, E. J. (2022).
AI in health and medicine.
Nature Medicine, 28(1), 31-38.

6. LeCun, Y., Bengio, Y., & Hinton, G. (2015).
Deep Learning.
Nature, 521(7553), 436–444.
https://doi.org/10.1038/nature14539
https://doi.org/10.1016/j.cell.2020.01.021